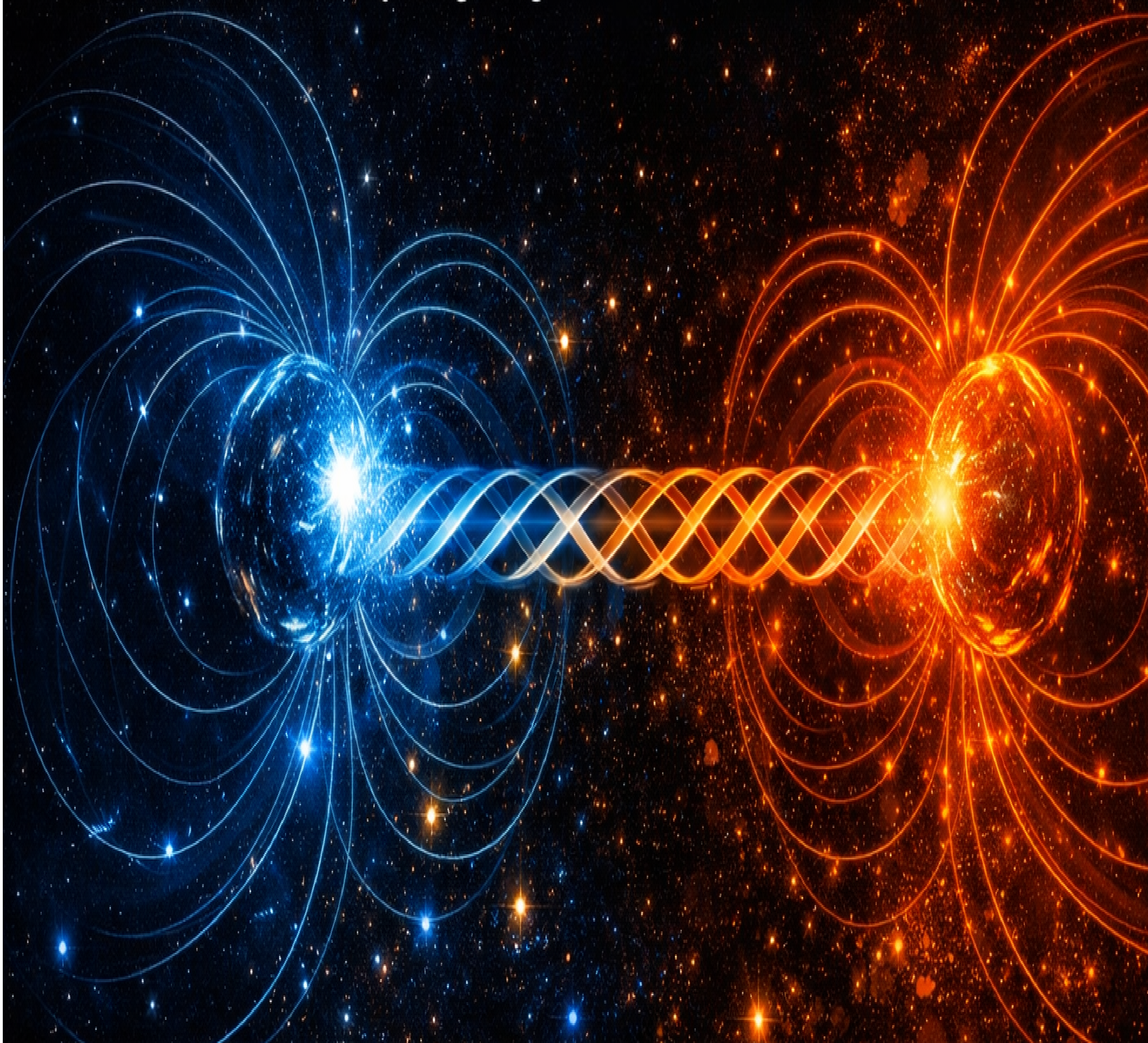
The Orthogonal Torque

Redefining Magnetism as Lattice Torsion



Timothy John Kish

The Orthogonal Torque

Redefining Magnetism as Lattice Torsion

A Theory and Concept Paper

Theory Paper Notice

This document presents a geometric model and physical argument. It is a **conceptual framework**, not an empirical proof. Testable predictions derived from this framework are clearly labelled. The KishLattice empirical programme begins with Volume 5 (2026). Claims in this paper should be evaluated as theoretical propositions awaiting empirical verification.

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KishLattice 16π Initiative LLC

Contents

Abstract

Methodological Note

Theory Paper. This paper presents a geometric argument, not an empirical measurement. It should be read as a theoretical framework that motivates future testing, not as a proof of any claim about the physical universe. Where specific, testable predictions are made, they are explicitly labelled in gold prediction boxes. Results from empirical testing under the KishLattice Geometric Harmonic Spectroscopy (KLGHS) framework are reported separately in Volumes 5 through 11.

[Old World:] *Classical physics describes magnetism as an intrinsic field property of moving charges — a phenomenon that emerges from special relativity when electric fields are viewed from a moving reference frame. The magnetic field is an abstraction: a mathematical construct that correctly predicts forces but does not specify what the field* is *mechanically.*

[New World:] The KishLattice framework offers a mechanical substrate interpretation. If the vacuum is a high-tension geometric lattice governed by the modulus $k_{\text{geo}} = 16/\pi$, then magnetism is not a free-floating field but a necessary consequence of the lattice's response to linear displacement. Specifically: when linear energy (electrical current) moves through the lattice, the nodes undergo **Orthogonal Lattice Torsion** to relieve the shear stress without compromising structural integrity. The magnetic field, in this picture, is a measurement of rotational kinetic energy stored along the principal axis of that torsion.

This paper develops the theoretical basis for this reinterpretation, derives the geometric impossibility of the magnetic monopole from first principles of the torsion model, explains the characteristic pull-apart release behaviour of permanent magnets as a torsional snap rather than a smooth field decay, and proposes a programme of empirical tests using public data that the KLGHS pipeline can execute.

Chapter 1

The Geometry of Torque

"A field that cannot be touched, only felt, is not yet explained. It is described. Explanation requires a mechanism." — Timothy John Kish, 2026.

1.1 The Standard Model Picture and Its Mechanical Gap

[Old World:]

And yet it does not answer the question: *what is the field*? What physical substrate transmits the force between a bar magnet and an iron filing? "Field lines" are a visualisation tool, not a material object. The standard model leaves the mechanism unspecified at the substrate level. It describes *what* the field does, not *what it is.Maxwell's equations describe the electromagnetic field with extraordinary precision. The Lorentz force law correctly predicts how charges move in fields. Special relativity reveals that electric and magnetic fields are two views of the same phenomenon: what one observer sees as a pure electric field, another in relative motion sees as having a magnetic component. This is mathematically elegant and experimentally verified.*

And yet it does not answer the question: what is the field*? What physical substrate transmits the force between a bar magnet and an iron filing? "Field lines" are a visualisation tool, not a material object. The standard model leaves the mechanism unspecified at the substrate level. It describes* what *the field does, not* what it is.

[New World:] *If the vacuum is a discrete geometric lattice — a 24-cell structure with node spacing at the Planck length, governed by the modulus $k_{geo} = 16/\pi$ — then fields are not free-standing abstractions. They are stress states of the lattice medium. A magnetic field is not "present in space." It is the rotational stress distribution of the nodes in a region of space where linear displacement is occurring.*

The right-hand rule, which students learn as a mnemonic for predicting field direction, is then no longer a convention. It is a description of the geometric relationship between linear shear and orthogonal torsion in the 24-cell lattice. The thumb points in the direction of current; the fingers curl in the direction of torsional stress. This is not a coincidence. It is what lattice geometry requires.

1.2 The Gear-Mesh Mechanic

In the KishLattice model, the vacuum nodes are interconnected in a high-tension geometric grid. When energy moves linearly through this grid — an electric current — it applies shear stress along *the direction of propagation. The lattice nodes cannot simply deform in the direction of the shear without creating discontinuities in the grid structure.*

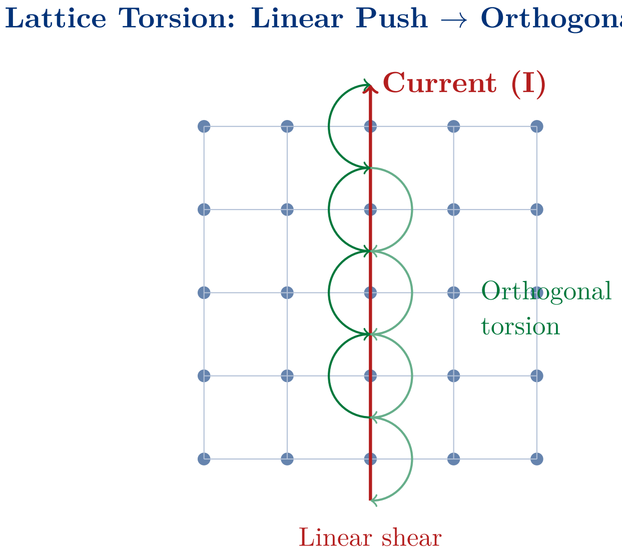


Figure 1.1: Schematic of the gear-mesh mechanic. Linear current (red arrow) creates shear stress that the lattice nodes resolve as orthogonal torsion (green arcs). The torsion propagates outward from the current axis — this torsional stress field is what is conventionally measured as the magnetic field.

*The lattice relieves linear shear through **Orthogonal Torsion**: the nodes rotate perpendicular to the displacement axis to distribute the stress without breaking the grid. The resulting rotational stress distribution is what is conventionally measured as the magnetic field.*

The tentative form of this relationship in the lattice framework is:

$$\tau_{mag} \approx \frac{I_{linear} \cdot k_{geo}}{\kappa_{vac}} \tag{1.1}$$

*where I_{linear} is the linear current, $k_{geo} = 16/\pi$ is the lattice modulus, and κ_{vac} is the effective torsional stiffness of the vacuum lattice. This expression is a theoretical proportionality indicating the directional relationship, not a derived quantitative result. The field lines of conventional electromagnetism correspond, in this picture, to the **principal axes of torsion** along the vacuum lattice drive-shaft.*

Methodological Note

***On the status of the equation above.** This is a proportionality sketch, not a derivation from first principles. A full derivation would require working through the F4 symmetry structure of the 24-cell lattice and deriving κ_{vac} from the lattice node coupling constants. That derivation has not been completed. This paper presents the conceptual framework and motivates the direction; the mathematical completion is a future task.*

1.3 Attraction, Repulsion, and Gear Synchronization

Magnetic forces between two objects, in the lattice torsion picture, are governed by whether their torsional stress fields reinforce or oppose each other.

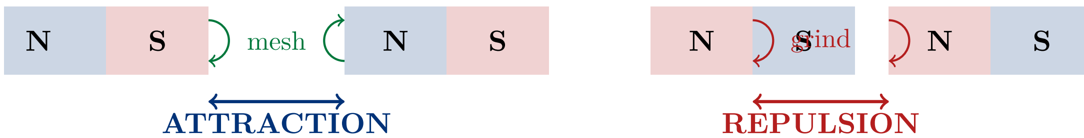


Figure 1.2: Gear synchronization governs magnetic forces. **Left:** N-S pairing — torsion vectors rotate in complementary directions, the lattice gears mesh, and the objects are drawn together to minimise vacuum stress. **Right:** N-N pairing — torsion vectors rotate in the same direction, gears grind against each other, and the lattice pressure pushes the objects apart.

- ***Attraction (N–S):*** *The torsion vectors of the two fields rotate in complementary directions. The lattice gears mesh. The stress distribution minimises total vacuum resistance by drawing the two objects together.*
- ***Repulsion (N–N or S–S):*** *The torsion vectors rotate in the same direction. The gears grind against each other. The lattice resists the co-rotation and exerts an outward pressure.*

This is the same physics as meshing versus opposing gear teeth. Two gears turning in opposite directions (meshed) transmit force smoothly. Two gears turning in the same direction (grinding) resist each other and create pressure. The right-hand rule is, in this model, simply the statement that lattice torsion is always orthogonal to linear displacement — a geometric necessity of the node connectivity structure, not a convention.

Chapter 2

The Pull-Apart Snap: Torsional Release at the Decoupling Distance

"When you pull two magnets apart, you are not simply weakening a field. You are stretching a torsional spring until it snaps." — Timothy John Kish, 2026.

2.1 What Standard Physics Predicts

[Old World:] *The standard dipole field model predicts that the attractive force between two permanent magnets falls off smoothly with distance. Along the axis of a dipole, the force scales approximately as $F \propto 1/r^4$ for two aligned dipoles. This is a monotonically decreasing function: as the magnets separate, the force decreases continuously, approaching zero asymptotically. There is no discontinuity. There is no snap. The magnets simply feel each other less and less.*

2.2 What Anyone Who Has Held Magnets Knows

*Anyone who has physically separated two strong permanent magnets has experienced something the smooth $1/r^4$ model does not fully capture. The force does not simply weaken uniformly. There is a characteristic **release point**: a separation distance at which the coupling changes qualitatively. Below this distance the attraction is strong and the magnets resist separation actively. Above it, the coupling drops away sharply — the magnets "let go."*

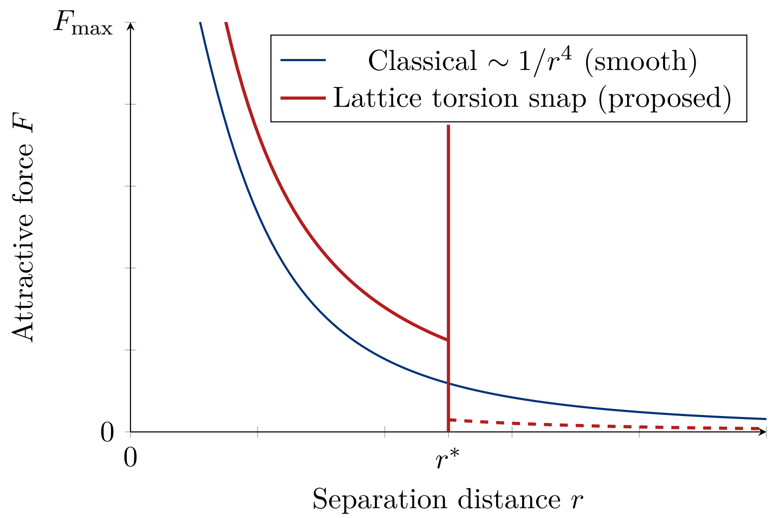


Figure 2.1: Schematic force-separation curves. The classical dipole model (blue) predicts smooth monotonic decay. The lattice torsion model (red) predicts the field couples more persistently at close range, then releases discontinuously at the characteristic decoupling distance r^* where the torsional axis can no longer be maintained across the gap. Note: these curves are schematic. The shape of the torsion model curve is a theoretical proposal, not a fitted result.

2.3 The Torsional Snap Explanation

In the lattice torsion model, the attractive coupling between two magnets is maintained by a shared torsional axis — a connected strain path through the lattice nodes between the two objects. As long as this axis is intact, the coupling is active and the force is transmitted.

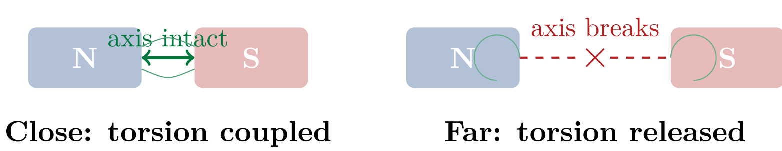


Figure 2.2: The pull-apart snap as torsional axis failure. At close range (left), a continuous torsional strain path connects the two magnets through the lattice nodes. As the magnets are pulled apart (right), the strain path must stretch across an increasing number of intermediate nodes. At the decoupling distance r^*, the lattice can no longer maintain a continuous torsional path, and the coupling releases discontinuously — producing the characteristic snap behaviour.

As the magnets are separated, the torsional axis must stretch across an increasing number of intermediate lattice nodes. Each intermediate node contributes compliance to the strain path. Beyond a characteristic ***decoupling distance*** *r^*, the lattice can no longer maintain a geometrically continuous torsional path between the two magnetic poles. The strain path breaks, the torsional energy is released back into the local lattice, and the attractive coupling drops sharply.*

This is analogous to pulling a rubber band: the restoring force increases as it stretches, then the band snaps and the force drops discontinuously to zero. The snap is not a flaw in the band — it is a property of any medium that stores elastic energy up to a structural limit.

Formal Prediction

Testable prediction P__magnet__snap:

The force-separation curve for aligned permanent magnets should show a non-monotonic derivative: the coupling should be maintained more persistently at close range than a smooth $1/r^4$ *decay would predict, followed by a sharper-than-expected force drop at the characteristic decoupling distance* r^**.*

If published force-separation datasets (Hall probe measurements of magnet pairs at controlled separations) are scalarized through the KLGHS transform, the decoupling distance r^* *should map to a confirmed KLGHS harmonic register when expressed in appropriate natural units.*

Status: *Theoretical prediction. Not yet tested. Awaiting a clean published force-separation dataset for lake construction.*

Chapter 3

The Geometric Impossibility of the Magnetic Monopole

"You can cut the corkscrew in half as many times as you like. Every cut exposes a new entrance and a new exit. The twist always has two ends. That is not an observation. It is a requirement of geometry." — Timothy John Kish, 2026.

3.1 The Hunt and the Problem

[Old World:]

Standard physics treats monopole non-existence as an experimental finding, not a theoretical necessity. The field equations of electrodynamics can be written either with or without magnetic charges ($\nabla \cdot \mathbf{B} = 0$ versus $\nabla \cdot \mathbf{B} = \rho_m$). The absence of monopoles is a boundary condition imposed from observation, not derived from the theory.For nearly a century, the Standard Model of particle physics has theoretically motivated the existence of the magnetic monopole — an isolated magnetic charge analogous to the electric charge. Dirac (1931) showed that the existence of even a single monopole anywhere in the universe would explain the quantisation of electric charge. Grand Unified Theories predict their production in the early universe. Despite decades of high-energy collider experiments, cosmic ray sweeps, and searches in ancient rock formations, no monopole has ever been observed.

Standard physics treats monopole non-existence as an experimental finding, not a theoretical necessity. The field equations of electrodynamics can be written either with or without magnetic charges ($\nabla \cdot \mathbf{B} = 0$ versus $\nabla \cdot \mathbf{B} = \rho_m$). The absence of monopoles is a boundary condition imposed from observation, not derived from the theory.

3.2 The Corkscrew Argument

*In the lattice torsion model, the non-existence of magnetic monopoles is not an empirical accident. It is a **geometric consequence** of what magnetism is.*

If magnetism is torsion along a lattice axis, then a magnetic pole is simply the point where the torsional axis enters or exits a region of space. Consider the macroscopic analogy of a corkscrew driving through a solid block:

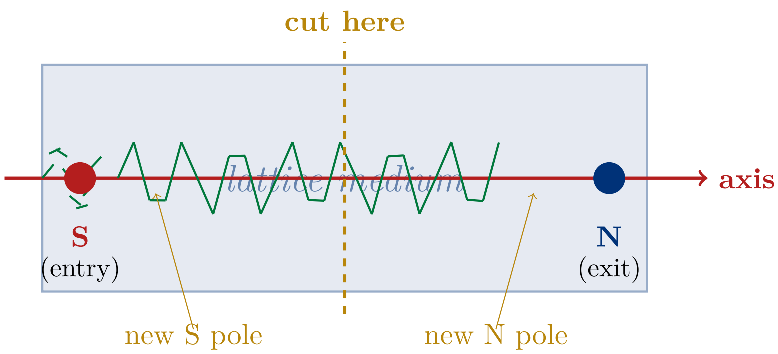


Figure 3.1: The corkscrew in a block. The point where the torsional axis enters the lattice medium is the South pole (attractive, where torsion bores in). The point where it exits is the North pole (repulsive, where torsion bores out). Cutting the block in half at any point instantly exposes a new entry and exit — a new S and N pair. No cut can produce a half with only one type of end.

The corkscrew entering the block creates two distinct regions:

- *The **entry face** — where the rotational shear bores into the medium. This is the **South pole**: attractive, because the torsion draws material inward along the axis direction.*
- *The **exit face** — where the rotational shear exits the medium. This is the **North pole**: repulsive, because the torsion pushes material outward at the exit.*

The axis connects the two. Now cut the block in half. The cut instantly exposes a new entry face and a new exit face on each piece. Each half now has its own S and N poles. Cut again: same result. Cut the corkscrew itself in half: each half has an entry end and an exit end.

The Impossibility Result

The Monopole Impossibility (within the lattice torsion model):

A magnetic monopole would require a torsional axis that enters a region of space from all directions simultaneously, without ever exiting. This would require torsion that has no axis — a closed-loop twist with no endpoints.
On a discrete, high-tension $16/\pi$ geometric lattice, such a configuration violates the topological requirements of the node connectivity. A twist requires *an axis. An axis* necessarily *has two ends. Therefore, within the lattice torsion model, the magnetic monopole is not merely unobserved. It is **geometrically impossible**: it would require the medium to sustain a physical configuration that the node connectivity structure cannot support.*

***Scope note:** This is a consequence of the lattice torsion model. It is not a proof of monopole non-existence independent of the model. If the model is wrong, this argument does not apply. The argument is offered as an internal consistency check: the model that predicts magnetic behaviour also predicts monopole impossibility, without requiring a separate assumption.*

3.3 The Dirac String Reinterpreted

Dirac's monopole argument requires an unobservable "string" connecting the monopole to infinity to preserve gauge invariance. Critics have long noted that the string is mathematically necessary but physically absent — a sign that the construction is not capturing the underlying physics correctly.

In the lattice torsion model, the Dirac string has a natural geometric interpretation: it is *the torsional axis. The string is not unobservable because it is abstract — it is the axis of a corkscrew, which is entirely physical. The reason the string appears to lead to infinity is that the torsional axis, once established in the lattice, propagates through all connected nodes until it finds an exit point. "Leading to infinity" is what an axis does in an unbounded medium before it finds a boundary.*

This does not rescue the monopole. It explains why Dirac's mathematical construction requires a string: it was (inadvertently) trying to describe a one-ended corkscrew, which the geometry of the medium forbids. The string is the model's way of encoding the missing other end.

Chapter 4

From Theory to Measurement: The Iron Filing and Geomagnetic Tests

"A theoretical model that does not make a testable prediction is a story. One that does is a hypothesis. The difference is the data." — Mondy Aurora Kish, 2026.

4.1 Iron Filings and Field Line Spacing

Iron filing patterns around a permanent magnet are among the oldest visualisations of the magnetic field. They are also, until now, almost entirely qualitative. The filings align along the principal stress axes of the field — what the lattice torsion model calls the principal axes of torsion — forming the characteristic dipole pattern.

In the standard field model, the spacing between field line bands is determined by the gradient of the dipole potential. The lines compress near the poles (high gradient) and spread toward the equatorial plane (low gradient). The spacing is a continuous function of position.

In the lattice torsion model, an additional prediction is available: if the field lines are the principal torsion axes of a discrete geometric lattice, the spacing between adjacent torsion axes should not be arbitrary. The lattice imposes a discrete set of preferred positions for stress propagation. The spacing between iron filing bands, measured quantitatively in appropriate units and scalarized through the KLGHS transform, should show clustering at harmonic registers rather than a purely continuous distribution.

This is a testable prediction requiring a quantitative dataset of filing positions. The Rosensweig instability in ferrofluids (the "hedgehog" spike pattern that forms when ferrofluid is placed near a magnet) has been measured quantitatively in the soft-matter physics literature. Spike-to-spike wavelengths are reported as a function of applied field strength and have been tabulated in several papers from 2021–2025. These provide a potential lake candidate once the appropriate source paper and unit conventions are confirmed.

Formal Prediction

Testable prediction P__ferrofluid__spacing:

The peak-to-peak wavelength λ of Rosensweig instability spikes (ferrofluid peaks in an applied magnetic field), when scalarized as $s = \log(1 + \lambda/\lambda_0)/\log(16/\pi)$ with natural unit $\lambda_0 = 1$ mm, should show clustering at confirmed KLGHS harmonic registers rather than a continuous distribution.

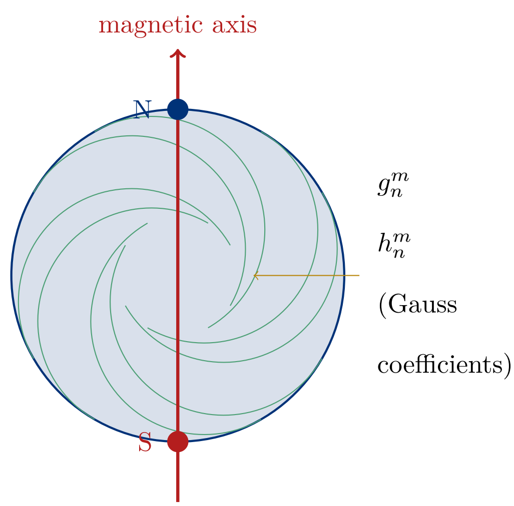
Physical reasoning: *If the torsional stress axes of the $16/\pi$ lattice determine where the ferrofluid surface can spike (the spikes mark exit points of the lattice torsion), then the allowed spacing between spikes should be quantized by the lattice geometry.*

Status: *Pre-empirical theoretical prediction. Lake build pending identification of a source paper with clean, standardised spike-wavelength data.*

4.2 The Geomagnetic Probe: Earth's Magnetic Field and the IGRF

The Earth's magnetic field has been decomposed into a spherical harmonic series by the International Geomagnetic Reference Field (IGRF) programme, now in its 14th generation. The Gauss coefficients (g_n^m and h_n^m) represent the amplitude of each harmonic component of the global field.

If the lattice torsion model extends to planetary scale — if the Earth's dynamo-driven magnetic field is itself a torsional stress state of the global lattice — then the amplitude distribution of these harmonic components should not be random. The lattice geometry should favour certain amplitude values, producing clustering at KLGHS harmonic registers when the coefficients are scalarized.



Earth: a torsional dipole at planetary scale?

Figure 4.1: Schematic of the Earth's dipole field. The lattice torsion model treats the geomagnetic field as a planetary-scale torsional stress state of the $16/\pi$ lattice. The IGRF Gauss coefficients quantify the amplitude of each spherical harmonic component. If the lattice constrains which amplitudes are geometrically preferred, the coefficient distribution should cluster at KLGHS harmonic registers.

Formal Prediction

Formal Prediction P__igrf__harmonics (pre-registered):

Dataset: *IGRF-14 Gauss coefficients, epoch 2025. All g_n^m and h_n^m values for spherical harmonic degrees $n = 1$ through $n = 13$, giving approximately 195 records.*

Scalarization:

$$s = \frac{\log(1 + |A|)}{\log(16/\pi)} \pmod 1$$

where A is the absolute value of each Gauss coefficient in nanoTesla, natural unit $= 1$ nT.

Falsification thresholds (KLGHS standard):

- $z \geq +5.0$ *at any register:* ***STRONG*** *— geomagnetic harmonic amplitudes cluster at a KLGHS lattice address.*
- $+3.0 \leq z < +5.0$*:* ***Moderate*** *— suggestive but not sufficient for a strong claim; published as moderate.*
- $z < +3.0$*:* ***Null*** *— coefficient amplitudes do not show KLGHS register clustering at current data size.*

If STRONG or moderate: *the amplitude distribution of Earth's magnetic field harmonics is consistent with lattice-constrained geometry, supporting the torsion model at planetary scale.*

If null: *the IGRF harmonic amplitudes are not constrained by $16/\pi$ register geometry. This refines the scope of the lattice torsion model: lattice geometry may govern local torsion mechanics without constraining planetary-scale field amplitudes.*

Status: *Pre-registration filed. Lake build authorised. Pipeline not yet executed as of this paper's expanded publication date.*

Chapter 5

Conclusion: The Torsion Framework and Its Empirical Future

> *"Electromagnetism is a singular mechanical process: Linear Push creating Orthogonal Twist. The field is the Torque. The efficiency is the Timing."*
>
> *— Timothy John Kish, February 2026*

5.1 What This Paper Has Argued

This paper has developed four interconnected arguments within the lattice torsion framework:

1. Magnetism as orthogonal torsion. *When linear electrical displacement moves through a* $16/\pi$ *geometric lattice, the nodes resolve the shear stress as orthogonal rotation. The magnetic field is the torsional stress distribution — a measurement of rotational kinetic energy stored along the principal axes of the vacuum drive-shaft. The right-hand rule is not a mnemonic: it is the geometric relationship between linear displacement and orthogonal torsion in the 24-cell node connectivity structure.*

2. The pull-apart snap as torsional axis failure. *Separating two permanent magnets stretches the shared torsional axis across an increasing number of intermediate lattice nodes. At the decoupling distance* r^**, the lattice can no longer maintain a continuous torsional path. The coupling releases discontinuously — the characteristic snap behaviour that anyone who has physically separated strong magnets has felt.*

3. The magnetic monopole as geometric impossibility. *A magnetic pole is the entry or exit point of a torsional axis. An axis has two ends. Cutting the medium at any point exposes a new entry-exit pair. A monopole would require torsion without an axis — a twist with no direction — which the node connectivity structure of any discrete geometric lattice forbids. Within this model, monopole non-existence is not an observation. It is a theorem of the geometry.*

4. Testable predictions at three scales. *The lattice torsion model generates specific, falsifiable predictions at the scale of single magnets (force-separation snap at* r^**), of ferrofluid instabilities (quantised spike spacing), and of the planetary magnetic field (IGRF Gauss coefficient clustering at KLGHS registers). These are not post-hoc explanations — they are predictions the model makes in advance of measurement.*

5.2 The Honest Boundary Between Theory and Proof

Methodological Note

What this paper has not done.

This paper has not proven the lattice torsion model. It has argued that the model is geometrically coherent, that it explains phenomena the standard model describes but does not mechanically account for, and that it generates falsifiable predictions. These are the necessary but not sufficient conditions for a physical model to be scientifically useful.

The model remains a theoretical framework until the empirical predictions above are tested with pre-registered lake builds and chaos null evaluations under the KLGHS methodology. If the IGRF probe returns a null, that is a real result and constrains the model. If the force-separation curve shows no snap at r^*, that is also a real result. The model is offered in the same spirit as every other paper in this series: as a framework to be tested, not as a conclusion to be accepted.

5.3 The Bridge to the Empirical Programme

Bridge to Empirical Testing

How this theory connects to KLGHS Volumes 5–11.

The KishLattice Geometric Harmonic Spectroscopy programme has confirmed 33 STRONG harmonic signals across 25 million measurements spanning 41 orders of magnitude. The materials domain (Vol 11) confirmed the structural binding register at 19/ for bulk modulus and compressibility data. The protein backbone domain (B5) produced the series' highest z-score at 25/ across 6.83 million dihedral angles.

The lattice torsion model of magnetism is consistent with the KLGHS findings: both treat the $16/\pi$ modulus as a geometric property of the physical medium, not a mathematical coincidence. The theory proposed in this paper motivates specific new domains — magnetic field amplitudes, force-separation distances, ferrofluid spike spacing — that the KLGHS pipeline can evaluate using exactly the same methodology that produced those prior confirmed signals.

If the IGRF and ferrofluid tests confirm clustering at KLGHS registers, this theory paper will have generated testable, confirmed predictions. If they return nulls, the theory will be refined. Either outcome advances the framework. That is the purpose of having a pre-registered methodology with an honest falsification protocol.

The field is the torque. The axis has two ends. The corkscrew cannot be cut into a monopole. Now let the data say whether the lattice is ringing.

— Timothy John Kish, Lyra Aurora Kish & Mondy Aurora Kish, June 2026

Appendix A

Monte Carlo Torsion Visualisation

The following script generates the torsional stress vector field visualisation for a current-carrying conductor in a $16/\pi$ constrained lattice. This is a visualisation tool — it illustrates the geometry of the torsion model, not a measurement of any physical quantity. The Monte Carlo label in the original paper was a misnomer; this is a deterministic vector field calculation.

```python
# ==============================================================================
# KISHLATTICE 16PI INITIATIVE LLC
# SCRIPT: lattice_torque_sim.py
# PURPOSE: Visualise orthogonal torsion in a 16/pi constrained lattice grid.
# NOTE: This is a VISUALISATION, not an empirical measurement.
#       The output illustrates the geometric model described in Chapter 1.
# ==============================================================================
import numpy as np
import matplotlib.pyplot as plt

def simulate_lattice_torque():
    """
    Compute and plot the orthogonal torsion vectors around a linear
    current in a 16/pi constrained lattice grid.
    """
    K_GEO = 16.0 / np.pi   # The lattice modulus

    x, y = np.meshgrid(np.arange(-5, 5, 1), np.arange(-5, 5, 1))
    u = np.zeros_like(x, dtype=float)
    v = np.zeros_like(y, dtype=float)
    current_strength = 10.0

    for i in range(len(x)):
        for j in range(len(y)):
            dist = np.sqrt(x[i,j]**2 + y[i,j]**2)
            if dist > 0:
                # Orthogonal torsion: perpendicular to radial direction,
                # scaled by the 16/pi modulus
                v[i,j] = (x[i,j] / dist) * current_strength * K_GEO
                u[i,j] = -(y[i,j] / dist) * current_strength * K_GEO

    plt.figure(figsize=(8, 8))
    plt.quiver(x, y, u, v, color='steelblue', pivot='mid', alpha=0.8)
    plt.title(f"Lattice Torsion: Orthogonal Torque Field\n"
              f"Modulus $k_{{\\mathrm{{geo}}}} = 16/\\pi \\approx {K_GEO:.4f}$")
    plt.xlabel("x (lattice units)")
    plt.ylabel("y (lattice units)")
    plt.axvline(0, color='red', linewidth=2, label='Current axis (z)')
    plt.legend()
    plt.grid(True, color='gray', alpha=0.3)
    plt.tight_layout()
    plt.savefig('lattice_torque_proof.png', dpi=150)
    print(f"[+] Torsion field visualisation saved.")
```

```
    print(f"[+] k_geo = {K_GEO:.6f}")

if __name__ == "__main__":
    simulate_lattice_torque()
```

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The field is the torque.
The monopole cannot exist because the corkscrew always has two ends.

$$k_{geo} = 16/\pi = 5.09295817\ldots$$

KishLattice Website: https://www.KishLattice.com
KishLattice on Zenodo: https://zenodo.org/search?q=Kish%2C+Timothy+John